

Sounak Chakraborty

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SUMMARY

MCA student (2027) at Christ University with experience building AI-powered applications, distributed backend systems, and full-stack web platforms using Java, Spring Boot, React, Django, PostgreSQL, Redis, and Docker. Interested in software engineering, backend development, and scalable systems.

EDUCATION

Master of Computer Applications (MCA) <i>Christ University, Bangalore</i>	2025 – 2027
Bachelor of Computer Applications (BCA) – CGPA: 7.8/10 <i>Techno Main Salt Lake, Kolkata</i>	2022 – 2025
Higher Secondary (Commerce) – 86% <i>South Point High School, Kolkata</i>	2022

PROJECTS

RepoSage – Scale-Safe Repository Intelligence System 2026

- Built a private offline codebase Q&A search engine using Django, Next.js 16 (React 19), PostgreSQL, and pgvector, supporting repositories with 10,000+ files.
- Implemented hybrid retrieval using HNSW vector search and full-text search, fused via Reciprocal Rank Fusion (RRF) and Cross-Encoder reranking, reducing irrelevant search results by approximately 35%.
- Designed an incremental ingestion pipeline with SHA-256 deduplication and checkpoint-resume processing, reducing re-indexing time by over 60% during repository updates.
- Built a telemetry console for monitoring execution latency and retrieval quality, alongside a staleness tracker for detecting code-to-document knowledge drift.

Gatekeeper – Intelligent Adaptive API Gateway 2026

- Engineered a self-healing API gateway using Node.js, Express, MongoDB, Redis, React, Socket.io, and Docker, sustaining over 5,000 requests per minute during load testing.
- Implemented Redis-backed token bucket rate limiting, three-state circuit breaking, and automated DDoS protection using configurable route-level thresholds.
- Built weighted round-robin load balancing, backend health monitoring, automatic failover, and W3C-compatible distributed tracing.

Fitoholic – Full-Stack AI-Powered Fitness Tracker 2025

- Built a fitness tracking web application using Spring Boot (Java 21), MySQL 8, and React 18 with real-time tracking of steps, calories, water intake, and weight.
- Integrated Razorpay for payment processing and Google Gemini API for AI-powered fitness recommendations, secured using Spring Security 6 and JWT authentication.
- Implemented interactive dashboards and animated visualizations using Framer Motion and Tailwind CSS to improve user engagement.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript, HTML, CSS

Frameworks & Libraries: Spring Boot, Django, React, Next.js, Express.js, Socket.io, Tailwind CSS

Databases & ORM: PostgreSQL (pgvector), MySQL 8, MongoDB, Redis, Spring Data JPA (Hibernate), Mongoose

Developer Tools: Git, Docker, Maven, Postman, VS Code, Eclipse

Core CS Concepts: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Backend Concepts: REST APIs, JWT Authentication, Spring Security, Circuit Breaker, Rate Limiting, Load Balancing, Distributed Tracing

AI & ML: RAG, Vector Search, Reciprocal Rank Fusion (RRF), Ollama, Google Gemini API

Operating Systems: Linux, UNIX, Windows

Integrations: Razorpay API, Google OAuth (NextAuth), SMTP Mail